

Interaction Obstructions and Resonant Breakdown of the Positive Pais–Uhlenbeck Metric

GPT-5.6.sol*

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Abstract

The free unequal-frequency Pais–Uhlenbeck (PU) oscillator admits a canonical positive pseudo-Hermitian metric, geometrically selected and classified in a companion series. This paper asks what happens to that metric — and to its Krein/ghost-parity alternative — when an interaction is switched on. For the oscillator with the cubic vertex inherited from a real z^3 interaction we construct the metric deformation explicitly through third order and prove: the deformation exists for generic frequencies; at the internal resonance $\omega_1 = 3\omega_2$ it is obstructed at second order by an exact on-shell conversion operator, and sufficiently excited resonant multiplets then acquire complex-conjugate energy pairs at $O(\zeta^2)$ — genuine spectral PT-breaking, which excludes *every* positive invariant metric where it persists; at $\omega_1 = \frac{3}{2}\omega_2$ a third-order obstruction appears; a transfer-lattice selection rule organizes the resonance hierarchy; and the deformation coefficients diverge geometrically, $R_n = O(\epsilon^{-3n/2})$, toward the Jordan boundary. For the perfect-square field theory $-\frac{1}{2} \int (\Box\phi + \lambda(\partial\phi)^2)^2$, regulated by a mass splitting, the cubic vertex is exactly the Källén polynomial $\frac{1}{2}\lambda_K(p_1^2, p_2^2, p_3^2)$; an even-ghost selection rule protects the positive metric at first order even above the ghost-decay threshold, while the sectorwise ghost parity is obstructed there by the real decay. At second order the momentum continuum turns the oscillator’s isolated resonances into generic scattering shells: the branch-changing process $H + L \rightarrow L + L$ obstructs the pointed-sector positive metric, with the exact value $401\sqrt{6}/39424$ at a rational kinematic point, hence on an open subset of the shell. An exact two-field rewriting, $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$, converts the theory into $-\partial U \cdot \partial V + \frac{\lambda^2}{2} U^2 V^2$ and exhibits the hidden ghost parity as the exact nonlinear involution $U \leftrightarrow V$, which exchanges two oppositely oriented null vacuum sectors and generates the $\Box^2$ Jordan dynamics about either of them; the regulated branch parity has no bounded confluent limit in one sector but converges on the doubled space to this exchange. All finite-dimensional and modewise identities are machine-verified.

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1 Introduction

A companion series [5, 6, 7, 8] analyzed the *free* fourth-order theories — the PU oscillator, the fourth-order scalar field, and linearized quadratic gravity — and established a three-level structure: one complex spectral covariance, several real forms (involutions), and several completions. The positive pseudo-Hermitian completion of Bender and Mannheim [1, 2] and the Krein/ghost-parity completion of Bateman and Turok [3] are real forms of one free theory, distinguished at Jordan degenerations.

This paper begins the interacting classification. The organizing object is the *deformation problem* for the canonical positive metric: given the free Dyson map ρ_0 with $h_0 = \rho_0 H_0 \rho_0^{-1}$ Hermitian, and a perturbation $H_\zeta = H_0 + \zeta V$, when does a deformed map $\rho_\zeta = \exp[-\frac{1}{2}(\zeta R_1 + \zeta^2 R_2 + \dots)]\rho_0$ keep $h_\zeta = \rho_\zeta H_\zeta \rho_\zeta^{-1}$ Hermitian? Order by order this is a cohomological question: the order- n equation is $[h_0, R_n] = S_n$ with a source built from lower orders, solvable iff the kernel projection $\Pi_{\ker \text{ad}_{h_0}} S_n$ — the *obstruction class* — vanishes. A parallel problem governs the Krein side: deform the ghost parity κ_0 so that $[\kappa_\zeta, H_\zeta] = 0$, $\kappa_\zeta^2 = 1$.

Results (oscillator). For the unequal-frequency PU oscillator with the cubic vertex $V = -iy^3$ inherited from a real z^3 interaction:

1. the first-order deformation exists for all $\omega_1 > \omega_2 > 0$, with an explicit Weyl-ordered cubic R_1 (Section 3);

2. the second-order deformation exists for generic (nonresonant) frequencies, with the assembled h_ζ Hermitian through $O(\zeta^2)$ (Section 4);
3. at the internal 3:1 resonance $\omega_1 = 3\omega_2$ the second-order class is nonzero and exactly $\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\omega_2^4} (a_1 a_2^\dagger{}^3 - a_1^\dagger a_2^3)$, the on-shell $1 \leftrightarrow 3$ conversion, unremovable by any first-order freedom (Section 5);
4. the obstruction is spectral, not merely metric: the $E = 27\omega_2$ resonant multiplet develops a complex-conjugate energy pair at $O(\zeta^2)$, verified both in degenerate perturbation theory and by exact diagonalization of the truncated Hamiltonian; the broken-PT region has width $|\omega_1 - 3\omega_2| = O(\zeta^2)$ (Section 6);
5. a third-order obstruction appears at $\omega_1 = \frac{3}{2}\omega_2$; a transfer-lattice and degree-parity selection rule organizes the candidate loci, and the verified pattern is that the coprime ratio $p:q$ first obstructs at order $p + q - 2$ when the mode-2 transfer p is odd (Section 7);
6. toward the Jordan boundary $\epsilon = \frac{1}{2}(\omega_1 - \omega_2) \rightarrow 0$ the deformation diverges geometrically, $R_n = O(\epsilon^{-3n/2})$ (Section 8).

Results (perfect-square field theory). For the interacting theory $S = -\frac{1}{2} \int (\Box\phi + \lambda(\partial\phi)^2)^2$, regulated by a mass splitting $\delta = m_1^2 - m_2^2 > 0$:

7. the symmetrized cubic vertex is exactly $\frac{1}{2}\lambda_K(p_1^2, p_2^2, p_3^2)$, the Källén polynomial; it vanishes on three massless shells and on one generalized Jordan leg, and couples Jordan modes in pairs (Section 9);
8. at first order an *even-ghost selection rule* makes the positive obstruction vanish identically — below and above the ghost-decay threshold $m_1 = 2m_2$ — while the sectorwise ghost parity $(-1)^{N_{\text{ghost}}}$ is obstructed above threshold by the real decay, with class proportional to $\lambda_K(m_1^2, m_2^2, m_2^2) = m_1^2(m_1^2 - 4m_2^2)$ (Section 10);
9. at second order the momentum continuum makes the resonances generic: the branch-changing scattering $H + L \rightarrow L + L$ is exactly energy conserving at open sets of momenta for *every* split mass pair, and the positive obstruction is nonzero there — with the exact value $401\sqrt{6}/39424$ at the rational center-of-mass point $m_H = \frac{3}{2}m_L$ (Section 12);
10. the theory admits an exact two-field rewriting $\mathcal{L} = -\partial U \cdot \partial V + \frac{\lambda^2}{2} U^2 V^2$ with $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$; the hidden ghost parity is the exact nonlinear involution $U \leftrightarrow V$, which exchanges the two null vacuum branches and generates the $\Box^2$ Jordan dynamics about either of them (Section 11);
11. the regulated branch parity has no bounded confluent limit inside one Jordan sector, but converges, on the doubling by the oppositely oriented sector, to the exact sector-exchange involution (Section 13).

The central statement.

Positive pseudo-Hermitian metrics can be deformed locally away from resonances, but interactions generate cohomological obstructions whenever branch-changing processes become exactly energy conserving. In finite systems these occur on isolated rational resonance loci and can produce complex spectra. In field theory the momentum continuum makes the corresponding scattering shells generic. The exact massless perfect-square theory nevertheless possesses a nonperturbative doubled involution — not as a parity within either pointed Jordan sector, but as an exchange between two oppositely oriented sectors.

Bookkeeping. Two expansion parameters must not be conflated. The *deformation order* is counted by a formal parameter ζ : $H(\zeta) = H_0 + \zeta H_3 + \zeta^2 H_4$, defining $R_1, R_2, \dots$ and the obstruction orders. The perfect-square coupling λ is a fixed parameter: in the two-field frame the Jordan quadratic structure and the vertex coefficients carry $g = \lambda^2$, while the nonlinear field map itself depends on λ . In the oscillator sections ζ multiplies the cubic vertex and coincides with the coupling.

Verification. Every finite-dimensional, modewise, and distributional identity in this paper is machine-verified: the first-order theory and second-order recursion (checks ID1–ID10), selection rules and third order (SR1–SR3, O3a–O3e), spectral PT-breaking and the Källén vertex (PT1–PT4, PS1–PS2 of `verify_pt_breaking.py`), the first-order field theory (PS-A–PS-H), the two-field rewriting and exact involution (TF1–TF7), the sectorwise second-order obstruction (SO1–SO7), and the exact kinematic point and confluent parity (HX1–HX3), in the repository accompanying the series (SymPy 1.14.0, NumPy 2.4.4, Python 3.14; release tag `paper5-v1.0` fixes the commit). Appendix A maps checks to theorems. Operator-domain statements (unbounded cubic interactions, formal perturbation series) are flagged where they occur.

Conventions. Oscillator conventions follow [5, 6]: normalized coordinates $V = (x, y, p, q)$, mode 1 = (x, p) at frequency ω_1 , mode 2 = (y, q) at ω_2 , positive normal form $h_0 = \frac{1}{2}[\omega_1\omega_2 p^2 + (\omega_1/\omega_2)x^2 + (\omega_2/\omega_1)q^2 + \omega_1\omega_2 y^2]$, rapidity $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$, $c = \cosh \frac{r}{2} = \omega_1/\sigma$, $s = \sinh \frac{r}{2} = \omega_2/\sigma$, $\sigma = \sqrt{\omega_1^2 - \omega_2^2}$. All operator computations use exact Weyl-symbol Moyal calculus (finite for polynomials; commutators with the quadratic h_0 reduce to $i \times$ Poisson bracket; cubic–cubic commutators include the Λ^3 term).

2 The deformation complex

Definition 2.1 (Deformation problem and obstruction classes). Let ρ_0 be the canonical free Dyson map, $h_0 = \rho_0 H_0 \rho_0^{-1} = h_0^\dagger$, and let $v = \rho_0 V \rho_0^{-1}$ be the transported vertex. Seek $\rho_\zeta = \exp[-\frac{1}{2} \sum_n \zeta^n R_n] \rho_0$ with $R_n = R_n^\dagger$ such that $h_\zeta = \rho_\zeta H_\zeta \rho_\zeta^{-1}$ is Hermitian order by order. The first two equations are

$$[h_0, R_1] = v^\dagger - v, \quad [h_0, R_2] = (v_2^\dagger - v_2) + \frac{1}{2}[R_1, v + v^\dagger], \quad (1)$$

where v_2 is the transported $O(\zeta^2)$ vertex when present. With $h_0 = \sum_E E P_E$, the equation $[h_0, R] = S$ has a Hermitian solution iff $P_E S P_E = 0$ for every E ; the obstruction class is $\mathfrak{o}_+ = \Pi_{\ker \text{ad}_{h_0}} S$, and the solution is $R = \sum_{E \neq E'} P_E S P_{E'} / (E - E')$ plus an arbitrary Hermitian

element of the commutant. The parallel Krein problem deforms the ghost parity: $[H_0, K_1] = [\kappa_0, V]$, $\{\kappa_0, K_1\} = 0$, with $\mathfrak{o}_K = \Pi_{\ker \text{ad}_{H_0}}[\kappa_0, V]$.

Anti-Hermitian (unitary) components of the generator are free and do not affect the Hermiticity conditions; additions to R_n from the commutant of h_0 are the usual metric ambiguity. “Gauge independence” of an obstruction class means invariance under both.

3 The cubic PU oscillator at first order

The real interaction z^3 continues, under the PU rotation $z = iy$, to $V = -iy^3$. The transported vertex is $v = -i(cy + isp)^3$ (the free Dyson map acts by $\rho_0 y \rho_0^{-1} = cy + isp$), whence

$$v^\dagger - v = 2ic(c^2y^3 - 3s^2yp^2). \quad (2)$$

Theorem 3.1 (First-order deformation). *For every $\omega_1 > \omega_2 > 0$ the first-order equation is solved by the Hermitian Weyl-ordered cubic*

$$R_1 = \mathcal{W} \left[\frac{2c^3}{\omega_1\omega_2} qy^2 + \frac{4c^3}{3\omega_1^3\omega_2} q^3 - \frac{6cs^2(2\omega_1^2 - \omega_2^2)}{\omega_1\omega_2(4\omega_1^2 - \omega_2^2)} p^2q + \frac{12cs^2\omega_1}{\omega_2(4\omega_1^2 - \omega_2^2)} pxy - \frac{12cs^2\omega_1}{\omega_2^3(4\omega_1^2 - \omega_2^2)} qx^2 \right], \quad (3)$$

with no singular denominator in the physical region ($4\omega_1^2 - \omega_2^2 > 0$); the equivalent Hermitian interaction is $h_1 = \frac{1}{2}(v + v^\dagger) = s(3c^2y^2p - s^2p^3)$. Near the Jordan boundary, $c \sim s \sim \sqrt{\omega}/(2\sqrt{\epsilon})$ and $R_1 = O(\epsilon^{-3/2})$, against the logarithmic divergence of the free generator Q_0 .

4 Second order: generic existence

Theorem 4.1 (Generic second-order deformation). *For incommensurate frequencies the second-order obstruction $\mathfrak{o}_+^{(2)} = \Pi_{\ker \text{ad}_{h_0}} \frac{1}{2}[R_1, v + v^\dagger]$ vanishes; the solved R_2 is Hermitian, and the assembled $h_\zeta = e^{-X/2} \star (h_0 + \zeta v) \star e^{X/2}$, $X = \zeta R_1 + \zeta^2 R_2$, is Hermitian through $O(\zeta^2)$ — verified end to end in exact Moyal calculus, including the $\frac{1}{8}[[h_0, R_1], R_1]$ term and the Λ^3 contribution that pure Poisson-bracket reasoning would miss. R_2 carries the resonance denominators $(\omega_1 - \omega_2)$ and $(\omega_1 - 3\omega_2)$; no others can vanish for $\omega_1 > \omega_2 > 0$.*

5 The exact 3:1 obstruction

Theorem 5.1 (Second-order obstruction at 3:1). *At $\omega_1 = 3\omega_2$ the first-order deformation still exists, but the second-order class is nonzero and exact:*

$$\mathfrak{o}_+^{(2)} = \frac{27\sqrt{3}}{320\omega_2^4} (a_1 a_2^{\dagger 3} - a_1^{\dagger} a_2^3), \quad (4)$$

the anti-Hermitian on-shell conversion of one mode-1 quantum into three mode-2 quanta ($\omega_1 = 3\omega_2$ makes the process exactly energy conserving). The class is gauge independent: it is unchanged under $R_1 \rightarrow R_1 + \{N_1, N_2, N_1 N_2\}$ -type commutant additions, under additions of the on-shell quintic kernel operators to R_2 , and under anti-Hermitian first-order generators — $\Pi_{\ker \text{ad}_{h_0}}[G, v + v^\dagger] = \Pi_{\ker \text{ad}_{h_0}}[v - v^\dagger, G] = 0$ for all such G .

There is therefore no analytic second-order deformation of the canonical positive metric at 3:1, within the Weyl-algebra class, continuously connected to the free metric. The physical meaning: the interaction contains a process that is exactly on shell, and changing the metric cannot rotate it away.

6 Spectral PT-breaking

The obstruction is not an artifact of the deformation scheme: it enters the spectrum.

Theorem 6.1 (Complex pair in the $E = 27\omega_2$ multiplet). *At $\omega_1 = 3\omega_2$ (set $\omega_2 = 1$), the second-order degenerate perturbation matrix on the ten-dimensional shell $\{|j, 27-3j\rangle\}_{j=0}^9$ is real tridiagonal with antisymmetric off-diagonal $K_{j,j+1} = -\frac{27\sqrt{3}}{640} \sqrt{(j+1)n(n-1)(n-2)}$, $n = 27-3j$, and explicit diagonal shifts. Its spectrum consists of eight real eigenvalues and one complex-conjugate pair*

$$\kappa_{\pm} = 2.44880738219938 \pm 0.224586593808125 i. \quad (5)$$

The pair is genuinely spectral: exact diagonalization of the truncated $h_0 + \zeta v$ at $\zeta = 0.02$ exhibits it with $\text{Im } E = \pm 0.2246 \zeta^2$ to 0.5%, stable across cutoffs. The lowest resonant doublet $\{|1, 0\rangle, |0, 3\rangle\}$ remains real: a nonzero metric obstruction does not automatically break the lowest pair.

Corollary 6.2 (Strength of the conclusion). *Wherever the complex pair persists in the interacting spectrum, no positive-definite invariant metric exists — analytic in ζ or not — since a positive metric would make h_{ζ} similar to a self-adjoint operator. (The statement is perturbative-spectral: the cubic interaction is unbounded and an operator-theoretic analytic-family argument would be needed for the untruncated theory.)*

Proposition 6.3 (Broken-PT tongues). *Detuning $\delta_r = \omega_1 - 3\omega_2$ adds $j \delta_r$ to the shell diagonal; the spectrum is complex for $|\delta_r| < \delta_c = 38.4151 \dots \times \zeta^2$ (exactly linear in ζ^2 ; the constant is multiplet-specific). An order- n resonance correspondingly produces an $O(\zeta^n)$ tongue.*

7 Third order and the selection rules

Theorem 7.1 (Transfer-lattice selection rule). *Grade monomials by energy transfer $(\Delta n_1, \Delta n_2)$, so $[h_0, M] = [(\Delta n_1)\omega_1 + (\Delta n_2)\omega_2] M$. Order- n objects have transfers in the n -fold sumset of the vertex transfer set, total degree $\leq n + 2$ (each commutator reduces degree by at least two), and degree parity $\equiv n \pmod{2}$. The order- n obstruction is therefore supported on the finite candidate sets: $\{2:1\}$ at order 1, $\{3:1\}$ at order 2, $\{3:2, 2:1, 4:1\}$ at order 3, with 5:1 first reachable at order 4. The lattice gives candidates; coefficients decide.*

Theorem 7.2 (Third order). *At generic frequencies the third-order obstruction vanishes and R_3 is Hermitian with odd total degrees ≤ 5 (confirming the parity rule). At $\omega_1 = \frac{3}{2}\omega_2$ the class is nonzero and exact,*

$$\mathfrak{o}_+^{(3)} = -\frac{117\sqrt{30}}{1120} i (a_1^2 a_2^{\dagger 3} + a_1^{\dagger 2} a_2^3), \quad (6)$$

the on-shell $2 \leftrightarrow 3$ conversion ($2\omega_1 = 3\omega_2$), gauge independent in the sense of Theorem 5.1. The candidates 2:1 (orders 1 and 3) and 4:1 (order 3) have vanishing coefficients.

Conjecture 7.3 (Resonance hierarchy). *The coprime ratio $\omega_1/\omega_2 = p:q$ first obstructs at order $p+q-2$ when the mode-2 transfer p is odd. (Verified: 3:1 at order 2, 3:2 at order 3. Predicted: 5:1 at order 4, 5:3 at order 6. Every obstruction found so far carries odd mode-2 transfer; the mechanism excluding even transfers is open.) The resonance set is an expanding hierarchy, not (so far) a dense set.*

8 The Jordan divergence hierarchy

Theorem 8.1 (Geometric divergence). *Near the Jordan boundary $\omega_{1,2} = \omega \pm \epsilon$,*

$$Q_0 = O(\log \epsilon^{-1}), \quad R_1 = O(\epsilon^{-3/2}), \quad R_2 = O(\epsilon^{-3}), \quad R_3 = O(\epsilon^{-9/2}), \quad (7)$$

the last three measured as clean decade exponents $(-1.50, -3.00, -4.49)$; the second order shows no small-denominator enhancement from $\omega_1 - \omega_2$ (those source components are suppressed).

Conjecture 8.2. $R_n = O(\epsilon^{-3n/2})$, *so the deformation's radius of convergence in ζ collapses as $\zeta_c \sim \epsilon^{3/2}$ toward the Jordan boundary. (Fixed-order scaling is established; a norm bound on the coefficients would upgrade this to a theorem.)*

9 The perfect-square vertex

The interacting theory is $S = -\frac{1}{2} \int (\Box \phi + \lambda(\partial \phi)^2)^2$ [3], with cubic vertex $V_3 \propto \Box \phi (\partial \phi)^2$ and quartic $V_4 \propto \frac{1}{2} (\partial \phi)^4$.

Proposition 9.1 (Källén form). *On $p_1 + p_2 + p_3 = 0$ the symmetrized cubic vertex is exactly*

$$\mathcal{V}_3 = p_1^2(p_2 \cdot p_3) + p_2^2(p_1 \cdot p_3) + p_3^2(p_1 \cdot p_2) = \frac{1}{2} \lambda_K(p_1^2, p_2^2, p_3^2), \quad (8)$$

with $\lambda_K(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2xz - 2yz$. Consequently: $\mathcal{V}_3$ vanishes on three ordinary massless shells; it vanishes with one generalized Jordan leg ($\nabla \lambda_K|_0 = 0$); it can be nonzero with two (Hessian entries 2, -2): the cubic vertex couples the Jordan sector in pairs. On the regulated decay channel, $\lambda_K(m_1^2, m_2^2, m_2^2) = m_1^2(m_1 - 2m_2)(m_1 + 2m_2)$: the vertex vanishes exactly at the threshold $m_1 = 2m_2$ and turns on continuously above it.

10 First order in the regulated field theory

Regulate by $P_\delta = (\Box + m_1^2)(\Box + m_2^2)$, $\delta = m_1^2 - m_2^2 > 0$. In the mode expansion every leg is on its branch shell, so on the kernel of ad_{h_0} the full four-momentum is conserved and the vertex coefficient is $\frac{1}{2} \lambda_K$ of the branch masses. The transported field mode, derived from the free Dyson transport and verified against the spectral Wightman function of [7], is

$$\phi'(k) = \frac{i}{\sqrt{2\delta}} \left[\frac{a_2 + a_2^\dagger}{\sqrt{\omega_2}} + \frac{a_1 - a_1^\dagger}{\sqrt{\omega_1}} \right], \quad \sigma = \sqrt{\omega_1^2 - \omega_2^2} = \sqrt{\delta} \text{ (k-independent)}. \quad (9)$$

Theorem 10.1 (First-order protection and Krein decay obstruction). *(i) Even-ghost selection rule: the transported reality factors make $v^\dagger - v$ vanish on every odd-ghost-count monomial. Even-ghost shells ($2 \rightarrow 22$, $11 \rightarrow 2$, ...) are kinematically closed for all $m_1 > m_2 > 0$. Hence $\mathfrak{o}_+^{(1)} = 0$ identically — below and above the decay threshold. (ii) The sectorwise ghost parity $\kappa_0 = (-1)^{N_{\text{ghost}}}$ has $\mathfrak{o}_K^{(1)} = 0$ below threshold, and above it a nonzero class supported on the open $1 \rightarrow 22$ shell, equal there to $\lambda_K(m_1^2, m_2^2, m_2^2)$ times the leg normalizations: a really decaying ghost has no conserved sectorwise parity, and the protection is exactly coextensive with $\lambda_K = 0$ (threshold and massless point). (iii) Scaling: at massive confluence both R_1 and K_1 diverge as $\delta^{-3/2}$; on massless Jordan paths $m_2^2 = \alpha\delta$ the positive R_1 scales as $\delta^{-1/2}$ with path-independent power (the selection rule removes the dangerous collinear term), while the Krein K_1 is $\delta^{-1/2}$ generically and $\delta^{-3/2}$ on the exceptional path $\alpha = 2/7$. (These are split-basis statements; the basis itself degenerates at $\delta = 0$, and a basis-independent formulation on confluent states is deferred.)*

11 The exact two-field rewriting

Theorem 11.1 (Two-field form and exact involution). *Introducing the auxiliary field ψ and the exact nonlinear change of variables $U = e^{\lambda\phi}$, $V = (\psi/\lambda)e^{-\lambda\phi}$ (unit pointwise Jacobian),*

$$\mathcal{L} = -\partial_\mu U \partial^\mu V + \frac{\lambda^2}{2} U^2 V^2 = -\frac{1}{2}(\partial X)^2 + \frac{1}{2}(\partial Y)^2 + \frac{\lambda^2}{8}(X^2 - Y^2)^2, \quad (10)$$

with $X, Y = (U \pm V)/\sqrt{2}$: the interaction is built from the $O(1,1)$ invariant. The exchange $U \leftrightarrow V$ is an exact symmetry; in the original variables it is the nonlinear involution

$$\kappa_0 : (\phi, \psi) \mapsto \left(\frac{1}{\lambda} \log \frac{\psi}{\lambda} - \phi, \psi \right), \quad (11)$$

invisible at any finite perturbative order in (ϕ, ψ) — geometrically, the transition map between the two charts $\phi_A = \frac{1}{\lambda} \log U$ and $\phi_B = \frac{1}{\lambda} \log V$ on the extended configuration space.

Proposition 11.2 (Null sectors and interaction-generated Jordan dynamics). *The vacuum manifold is $UV = 0$: two null branches. The perturbative vacuum $(U, V) = (1, 0)$ is a nonzero null background; κ_0 maps it to $(0, 1)$ — it exchanges pointed sectors and is not a particle parity within either. Expanding about $(1, 0)$:*

$$\mathcal{L}^{(2)} = -\partial u \partial v + \frac{g}{2} v^2, \quad \mathcal{L}^{(3)} = g u v^2, \quad \mathcal{L}^{(4)} = \frac{g}{2} u^2 v^2, \quad g = \lambda^2, \quad (12)$$

with equations $\square v = 0$, $\square u = gv$: the exact $\square^2$ Jordan pair, the off-diagonal supplied by the interaction coupling. About $(0, 1)$ the mirror expansion has $u \leftrightarrow v$: the oppositely oriented chain. A chain-preserving involution commuting with a single Jordan block is $\pm I$ (companion lemma), so the exchange structure is the only possible realization of ghost parity — as the exact κ_0 indeed exhibits. The interactions in this frame are pure potentials: $H_{\text{int}} = -L_{\text{int}}$ exactly, with no Ostrogradsky–Legendre corrections (in the (ϕ, ψ) frame the quartic contains time derivatives and the canonical second-order vertex differs from $-L_4$).

12 The sectorwise second-order obstruction

Regulate the sector- A theory by $\mathcal{L}_0 = -\partial u \partial v - \mu^2 uv + \frac{g}{2}v^2 + \frac{\epsilon}{2}u^2$ (masses $m_\pm^2 = \mu^2 \pm \sqrt{\epsilon g}$; the regulator necessarily breaks κ_0), quantize in the positive frame via the branch eigenvectors ($v = \rho_b u$ on branch b , $\rho_\pm = \mp \delta/2$), and use the exact matrix-element form of the second-order source:

$$\langle \text{out} | \mathcal{S}_+^{(2)} | \text{in} \rangle = \langle v_2^\dagger - v_2 \rangle + \sum_n \frac{\langle \text{out} | v_1^\dagger | n \rangle \langle n | v_1^\dagger | \text{in} \rangle - \langle \text{out} | v_1 | n \rangle \langle n | v_1 | \text{in} \rangle}{E - E_n}, \quad (13)$$

which follows from $(v^\dagger - v)(v + v^\dagger) + (v + v^\dagger)(v^\dagger - v) = 2(v^\dagger v^\dagger - vv)$.

Theorem 12.1 (Generic sectorwise obstruction). *For the branch-changing scattering $H + L \rightarrow L + L$ — kinematically open at suitable momenta for every $m_1 > m_2$ — the source element (13) is nonzero. At the rational center-of-mass point $m_H = \frac{3}{2}m_L$ (both incoming particles at rest, outgoing momenta $\pm \frac{3}{4}m_L$, all shell conditions exactly satisfied) its exact value is*

$$\boxed{\mathcal{M}_{\text{obs}} = \frac{401\sqrt{6}}{39424}} \quad (14)$$

(with $m_L = 4$ in lattice units), the quartic contact piece vanishing exactly. Since the source is analytic on the nonsingular part of the shell, the obstruction is nonzero on an open subset: the second-order positive deformation of the pointed sector is generically obstructed. The computation is truncation-complete (tree-level exchange: the s, t, u channels close on the momentum set; verified by cutoff independence) and reproduced at three independent tuned kinematic points.

Remark 12.2 (Oscillator versus field theory).

PU oscillator	field theory
discrete energy spectrum	continuous momentum spectrum
isolated rational resonance	generic scattering shell
3:1 first appears at order 2	$H + L \rightarrow L + L$ appears at order 2
finite resonant multiplet	continuum of degenerate in/out states

The continuum turns the oscillator's discrete resonance obstruction into a generic scattering-shell obstruction. The exact doubled κ_0 does not cure it: it maps the obstructed sector- A construction to the (equally obstructed) mirror sector- B construction.

Remark 12.3 (Strength of the field-theory statement). Theorem 12.1 establishes a perturbative cohomological obstruction: there is no analytic second-order deformation of the canonical positive metric, in the specified regulated Fock representation and operator class, for a pointed sector. Unlike the oscillator case (Theorem 6.1), we have not yet exhibited complex spectra here; the analogous upgrades would be complex finite-volume levels, non-real resonance poles, or failure of pseudo-unitarity of the pointed-sector S -matrix. The distinction — analytic metric obstruction versus absolute nonexistence of all positive metrics — is maintained throughout.

13 The confluent parity theorem

Theorem 13.1 (Confluent limit of the regulated parity). *Let $P_\delta b_\pm = \pm b_\pm$ be the regulated branch ghost parity and $c = (b_+ + b_-)/2$, $d = (b_+ - b_-)/\delta$ the confluent Jordan basis. Then*

$$P_\delta = \begin{pmatrix} 0 & \delta/2 \\ 2/\delta & 0 \end{pmatrix} \quad \text{in the } (c, d) \text{ basis:} \quad (15)$$

$P_\delta^2 = 1$ but P_δ has no bounded limit as $\delta \rightarrow 0$ — the regulated ghost parity cannot converge as a chain-preserving parity inside one Jordan sector. On the doubled space with the oppositely oriented confluent identification in the mirror sector ($c_B = (b_+ - b_-)/2$, $d_B = (b_+ + b_-)/\delta$), the cross parity $b_\pm^A \leftrightarrow \pm b_\pm^B$ equals, exactly and δ -independently, the sector exchange $(c_A, d_A) \leftrightarrow (c_B, d_B)$. The regulated branch parity thus converges, after doubling by the oppositely oriented Jordan sector, to the exact involution $U \leftrightarrow V$.

This connects the split Krein algebra directly to the exact nonlinear symmetry, and explains the earlier observations from one statement: the sectorwise $(-1)^{N_{\text{ghost}}}$ is broken by real decay and has no confluent limit, while the exact parity survives — on the doubled state space.

14 Discussion

The phase picture. Away from resonances the positive pseudo-Hermitian metric deforms locally, at first and second order, with explicit generators. The failures are structural, not accidental: (i) on-shell branch-changing processes produce gauge-independent cohomological obstructions — isolated rational loci for the oscillator, generic scattering shells in the field theory; (ii) at the oscillator resonances the obstruction is spectral, producing complex-conjugate pairs in excited multiplets and $O(\zeta^2)$ broken-PT tongues; (iii) independent of resonances, the deformation loses uniformity toward Jordan degeneration, $R_n = O(\epsilon^{-3n/2})$. The Krein side is not simply the survivor: its sectorwise parity is broken by real ghost decay above threshold. What survives is subtler — an exact, nonperturbative, sector-exchanging involution of the massless perfect-square theory, which the confluent parity theorem identifies as the $\delta \rightarrow 0$ limit of the regulated branch parity on the doubled state space.

Symmetry enhancement at the boundary. The massless perfect-square point is not the continuous limit of the split theory with parameters set to zero: the Källén vertex degenerates there, the dangerous cubic amplitudes vanish, the doubled $O(1, 1)$ structure emerges, and the Jordan dynamics is generated by the interaction about its own null vacua. This mirrors the conformal boundary of the gravity companion [8], where a Weyl gauge symmetry emerges at a singular locus and the boundary theory must be quotiented and completed differently.

Superselection (open). Whether the two pointed sectors admit cat-like parity vacua $|0_\pm\rangle = (|0_A\rangle \pm |0_B\rangle)/\sqrt{2}$, or are superselected in infinite volume (spontaneously broken κ_0 in each pointed completion), requires finite-volume overlap and zero-mode analysis: the vacuum manifold is null, the kinetic form indefinite, and the original chart contains only one branch. The present results are stated so as not to depend on the resolution.

Predictions and next steps. The hierarchy conjecture predicts a 5:1 obstruction at fourth order (an $O(\zeta^4)$ tongue); only the kernel projection at $\omega_1 = 5\omega_2$ is needed to test it. For the field theory, the basis-independent confluent-state evaluation of R_1 will decide whether the $\delta^{-1/2}$ split-basis divergence is physical. The gravitational extension — the cubic Einstein–Weyl vertices hhM , hMM , MMM against the covariant real forms of [8] — is the natural target after that; the present results sharpen its expected shape: pointed positive completions obstructed by ghost-branch-changing graviton processes, with any surviving structure living on a doubled completion.

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A Verification checks and the theorems they support

Scripts (directory `symbolic/` of the repository), each printing ALL PASS:

Check	Supports
ID1–ID5 (<code>verify_interaction_deformation.py</code>)	Theorem 3.1
ID6–ID8	Theorem 4.1
ID9a–b	Theorem 5.1
ID10	Theorem 8.1 (R_2)
SR1–SR3 (<code>verify_interaction_order3.py</code>)	Theorem 7.1
O3a–O3d	Theorem 7.2
O3e	Theorem 8.1 (R_3)
PT1–PT3 (<code>verify_pt_breaking.py</code>)	Theorem 6.1
PT4	broken-PT tongues
PS1–PS2	Proposition 9.1
PS-A–PS-G (<code>verify_perfect_square.py</code>)	Theorem 10.1
PS-H	Jordan-chain lemma used in Proposition 11.2
TF1–TF7 (<code>verify_two_field.py</code>)	Theorem 11.1 , Proposition 11.2
SO1–SO7 (<code>verify_sector_obstruction.py</code>)	Theorem 12.1 (numerics)
HX1 (<code>verify_hardenig.py</code>)	Theorem 12.1 (exact value)
HX2–HX3	Theorem 13.1